

# Advanced Digital Systems Design

## Lab 2

### Part 1

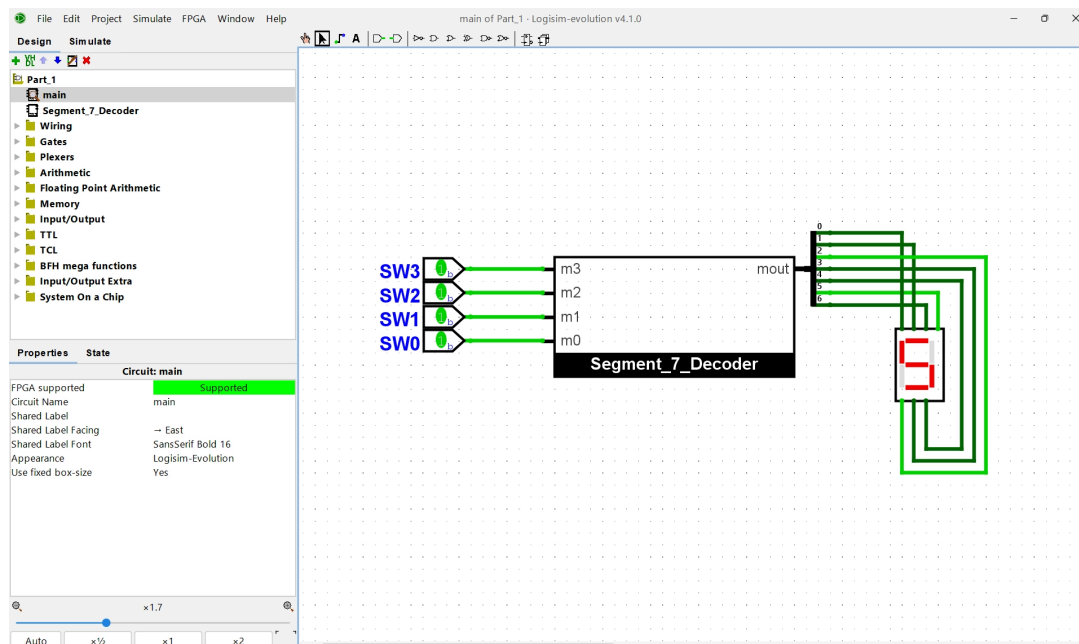


Figure 1: Logisim Circuit of Seven Segment Decoder. Mine would wrap around once more than 9.

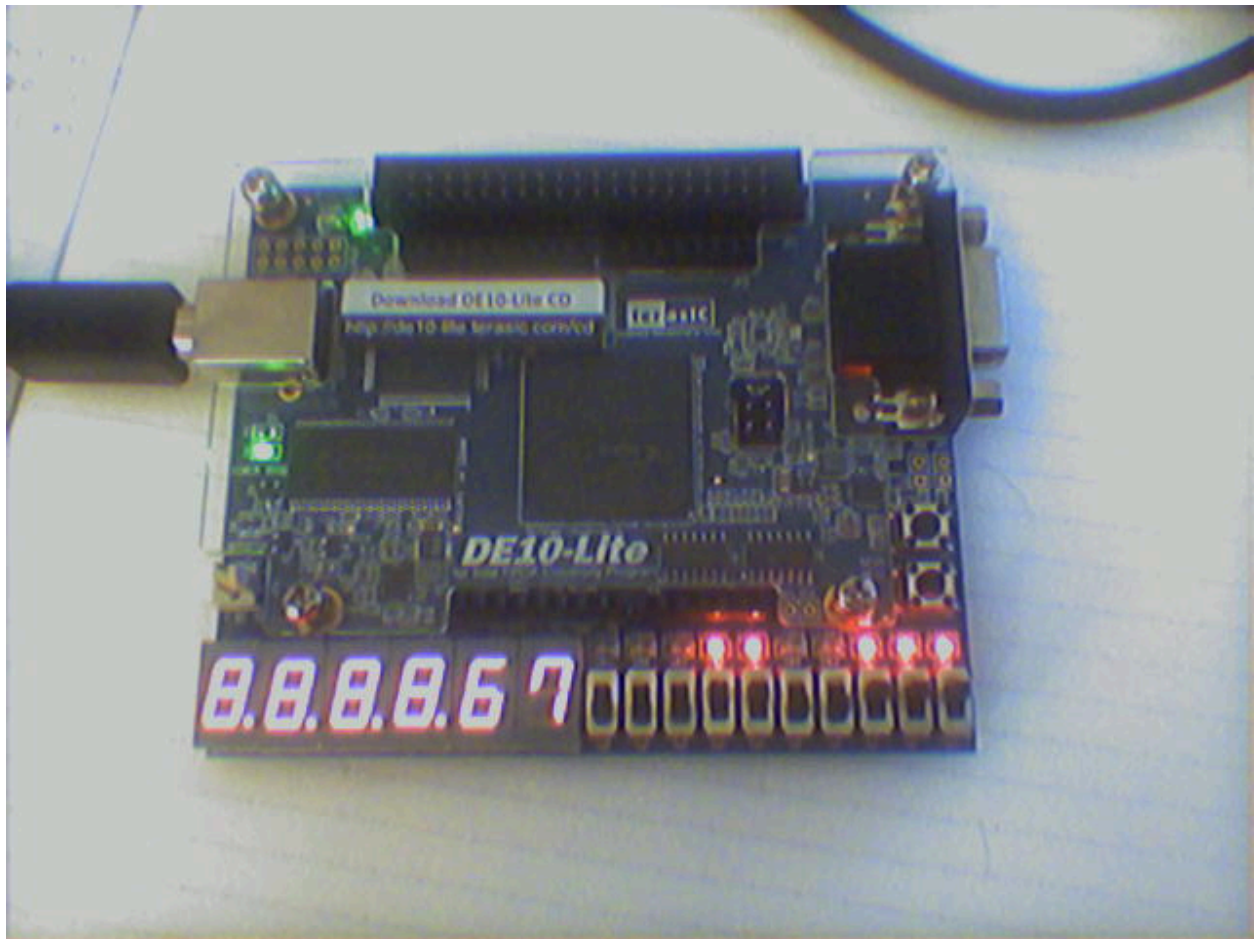


Figure 2: Part 1 Board

## Part 2

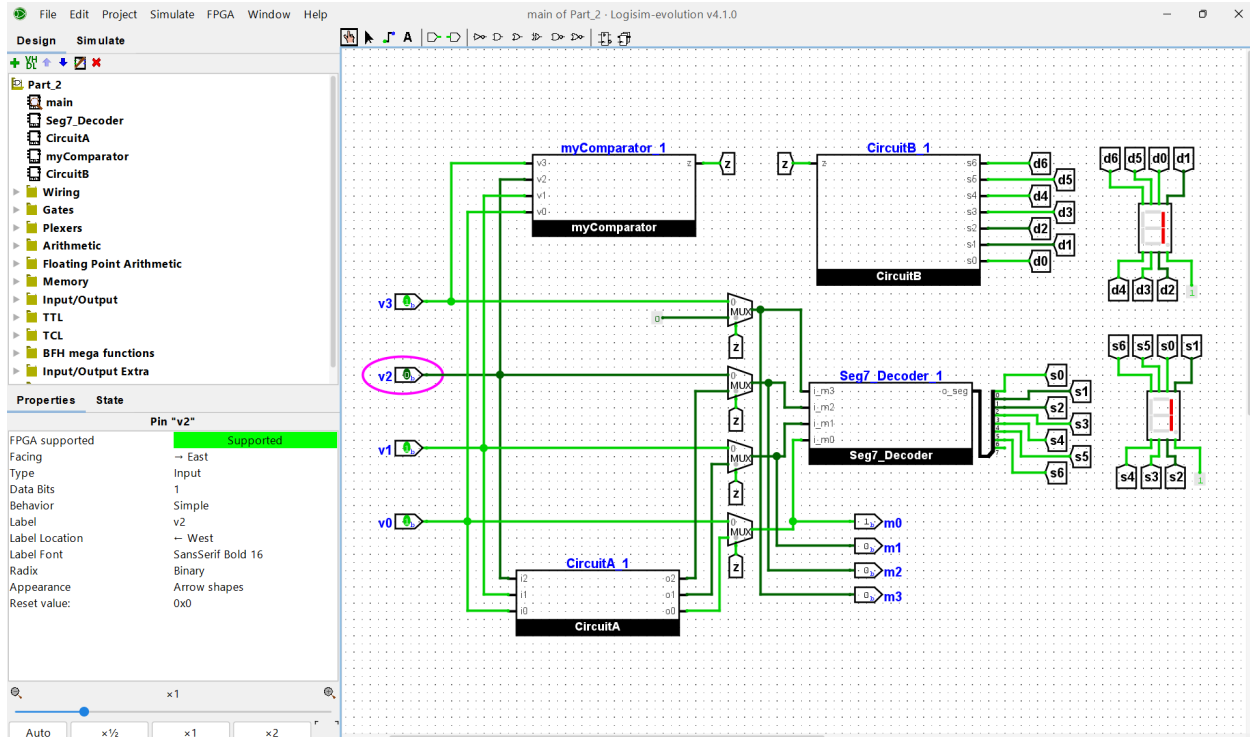


Figure 3: Logisim Circuit for Part 2

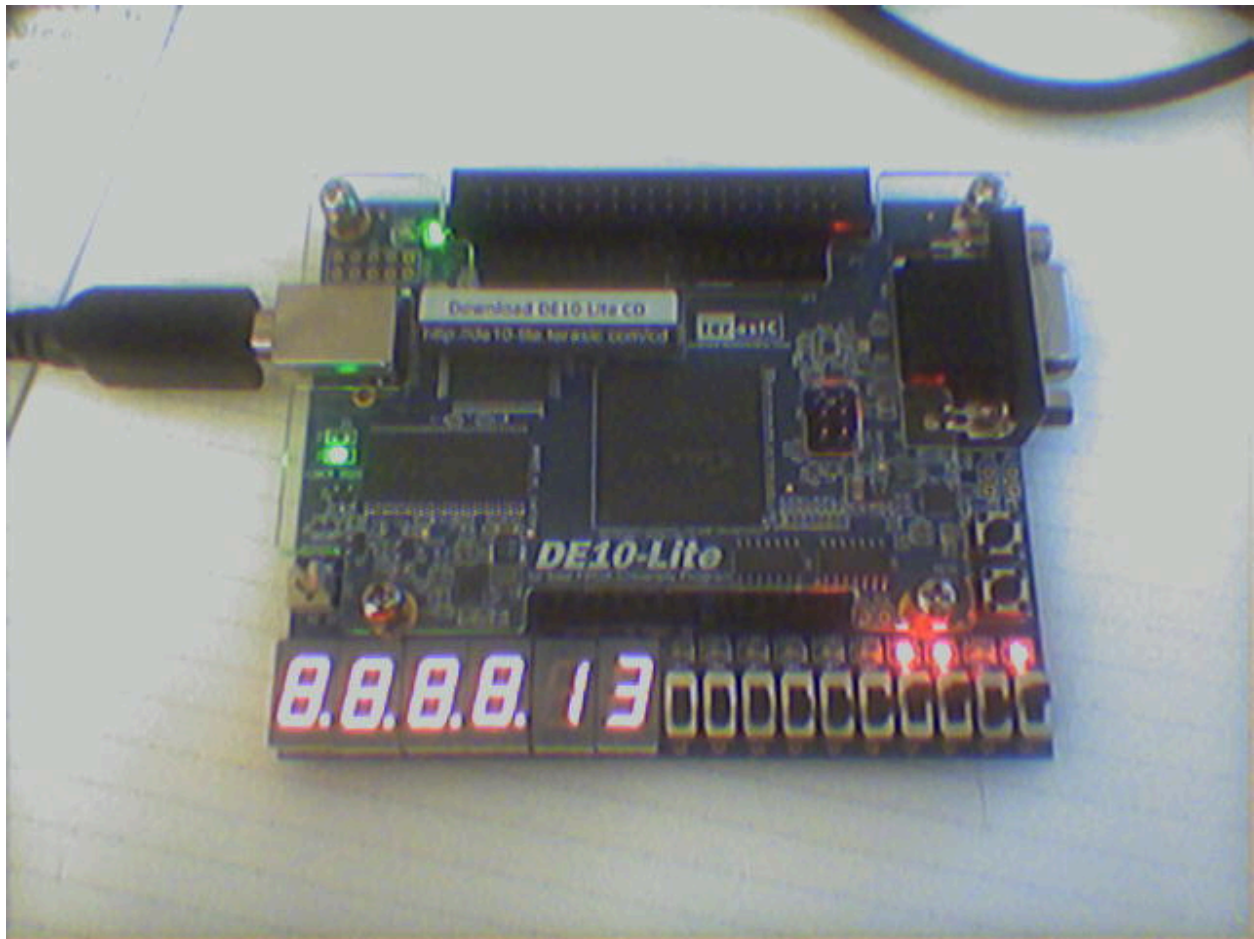


Figure 4: Part 2 Board

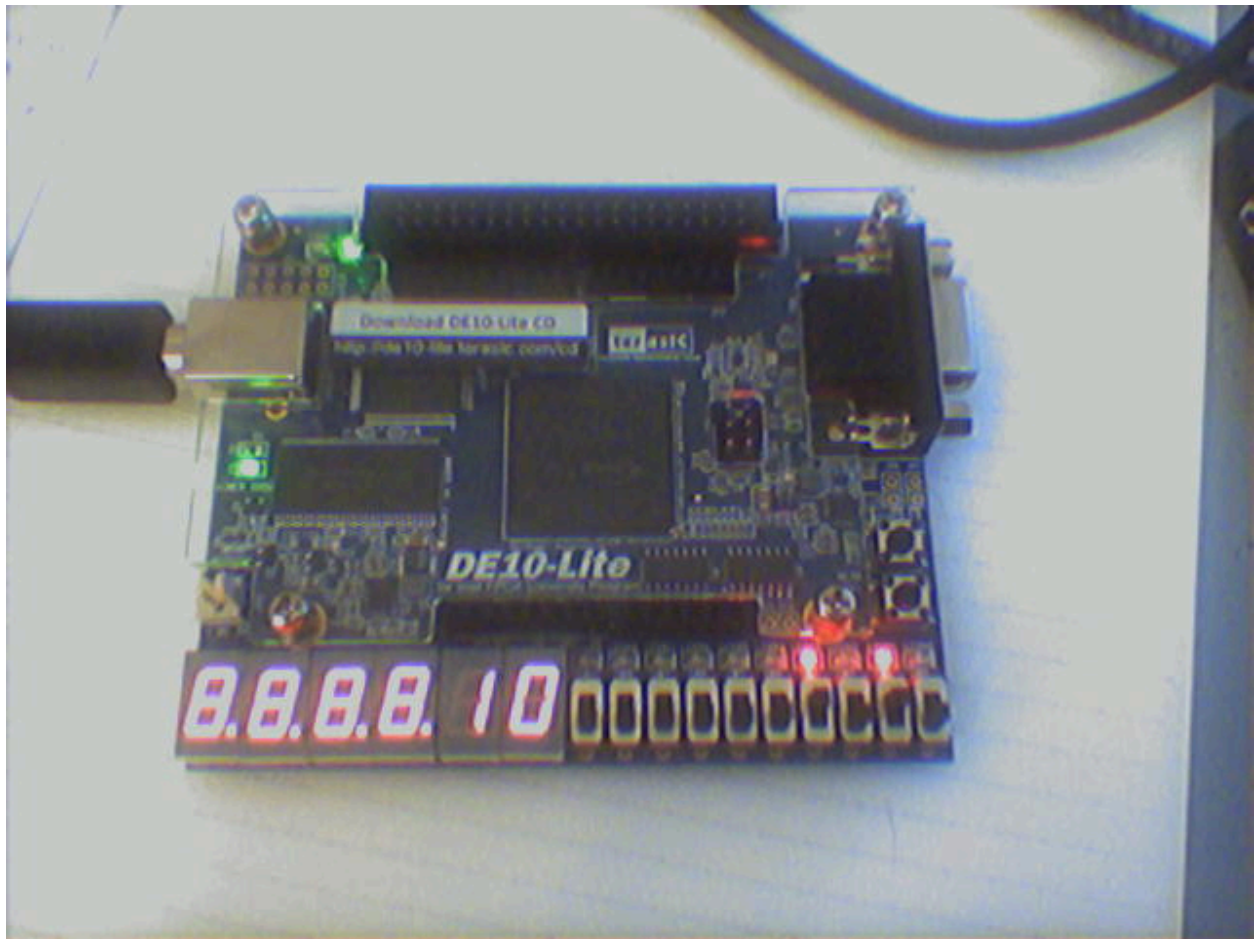


Figure 5: Part 2 Board

## Part 3

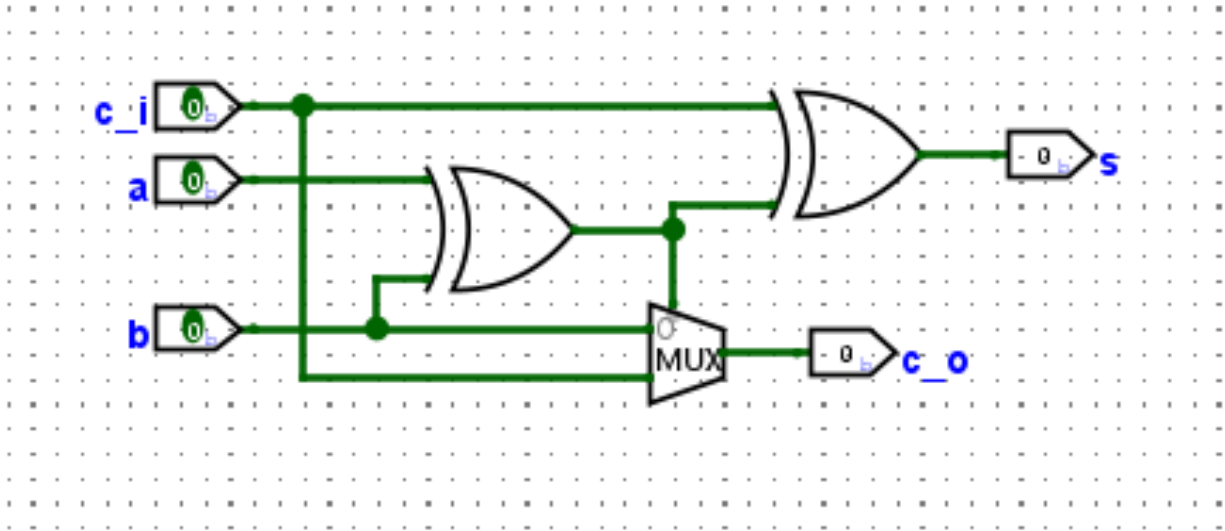


Figure 6: Logisim Full Adder

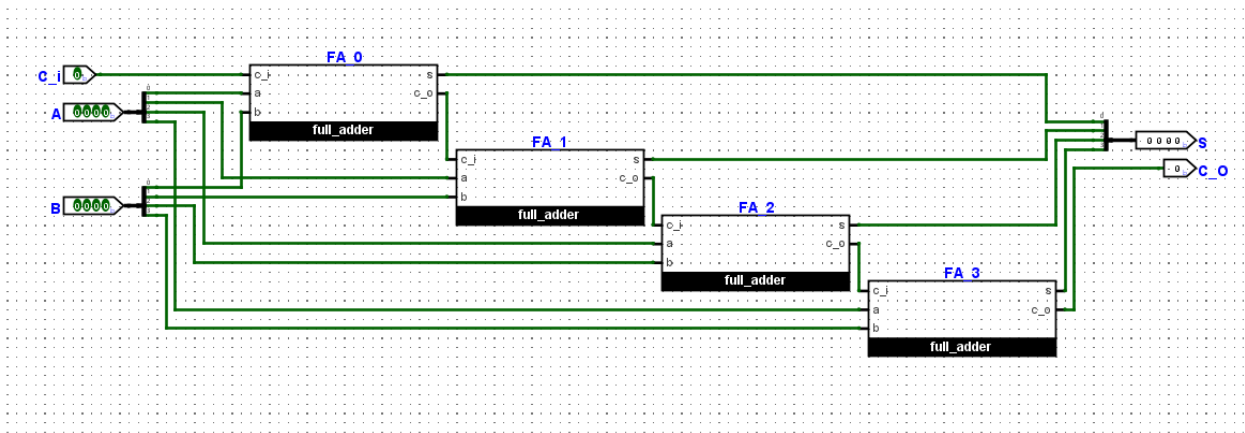


Figure 7: Logisim 4 Bit Adder

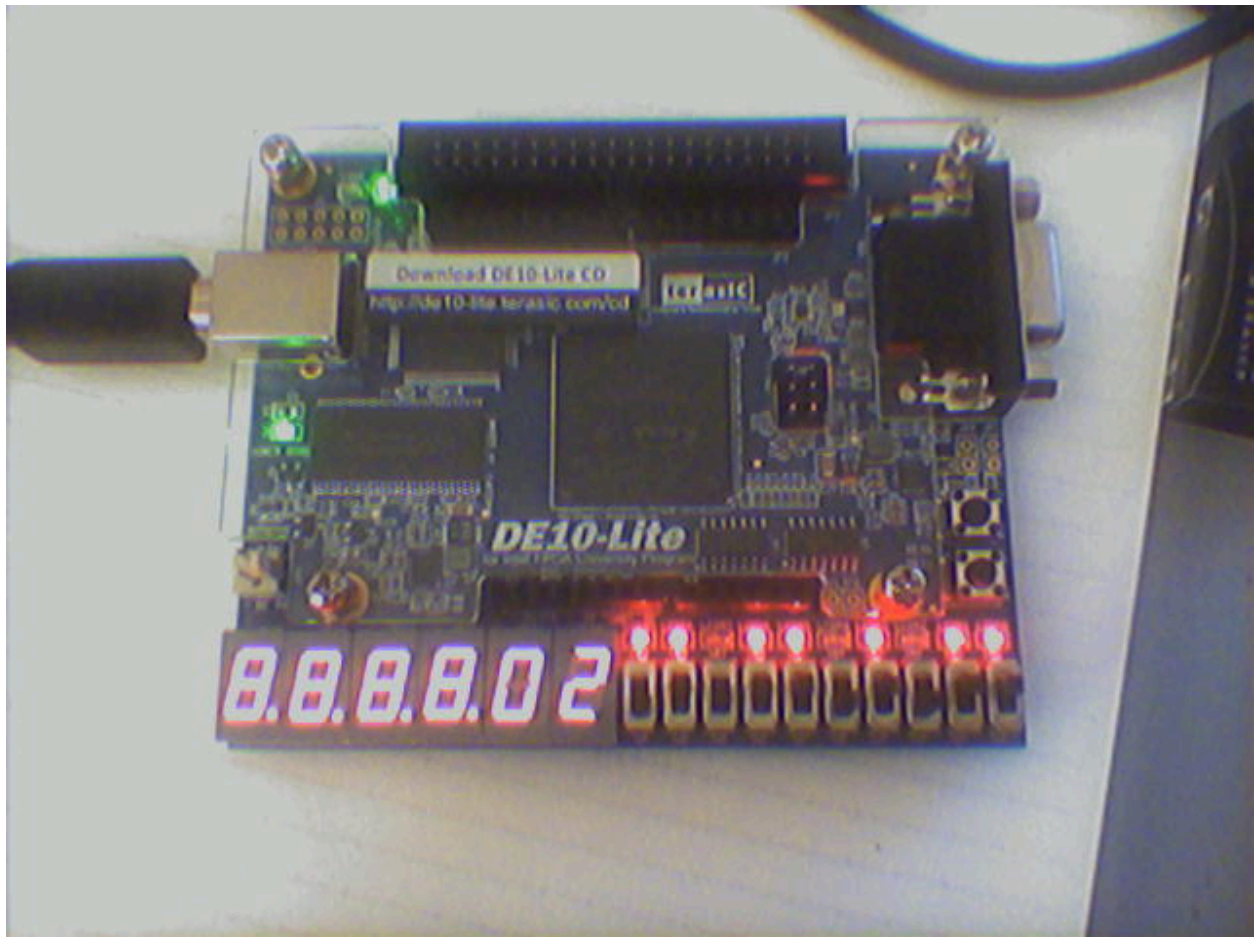


Figure 8: Part 3

## Part 4

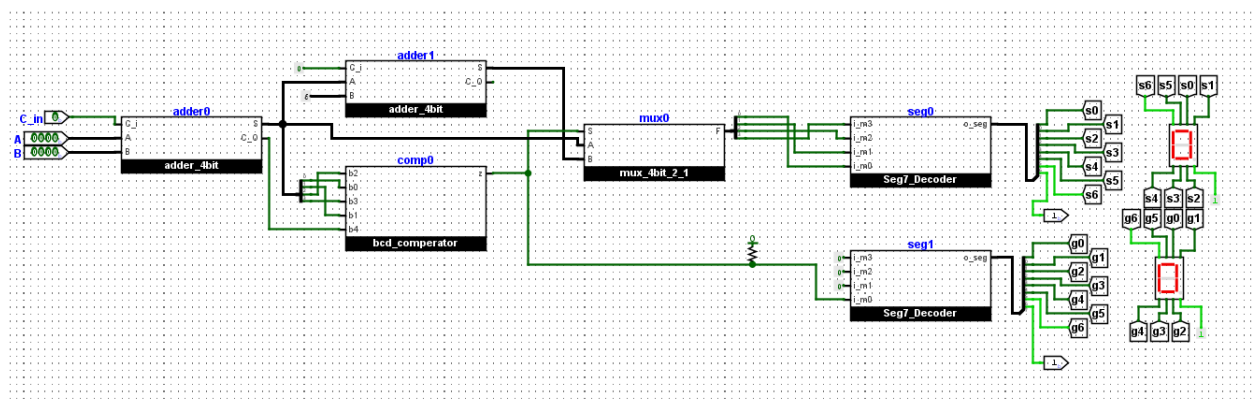


Figure 9: Logisim BCD Adder

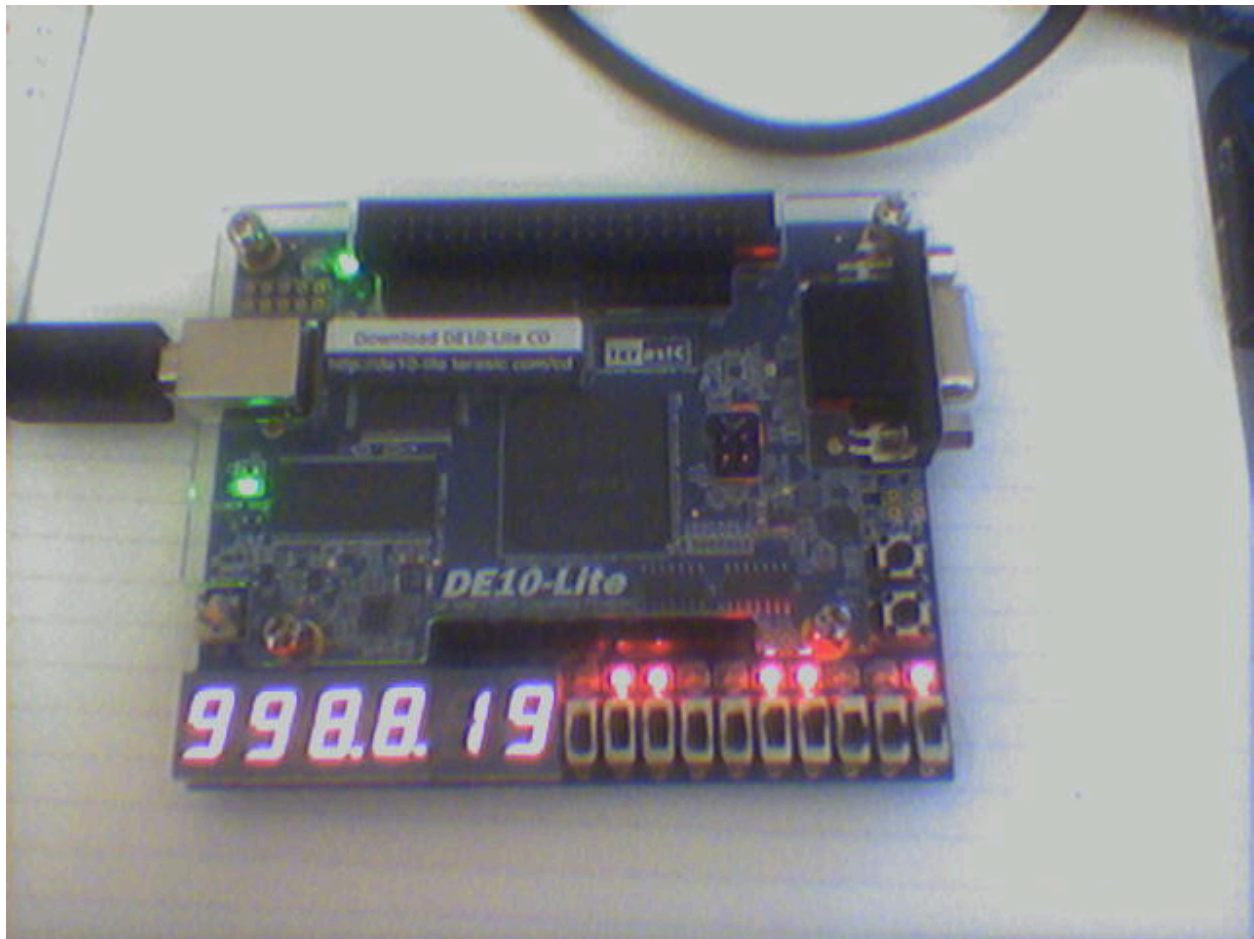


Figure 10: Part 4 Board

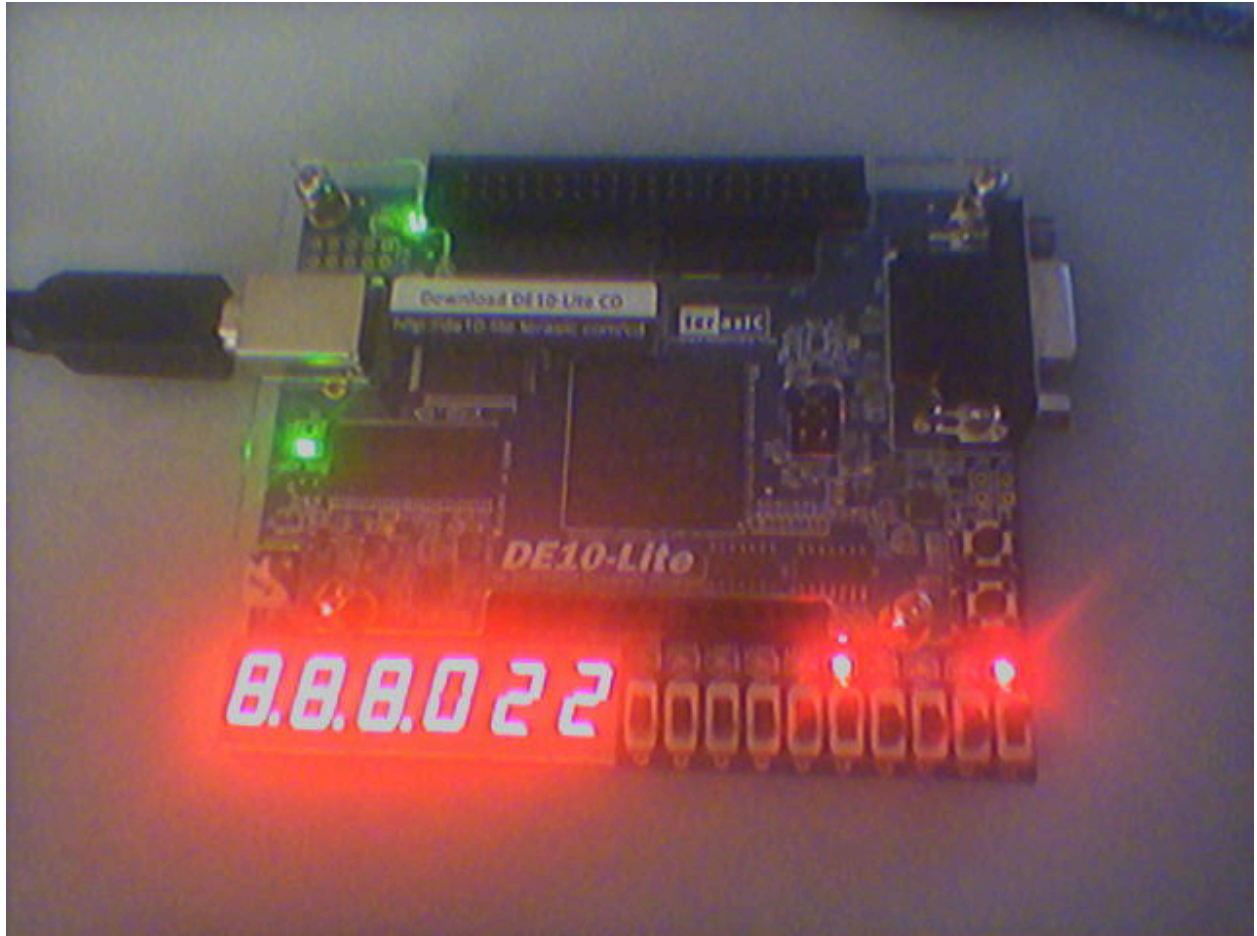


Figure 11: Part 4 Board

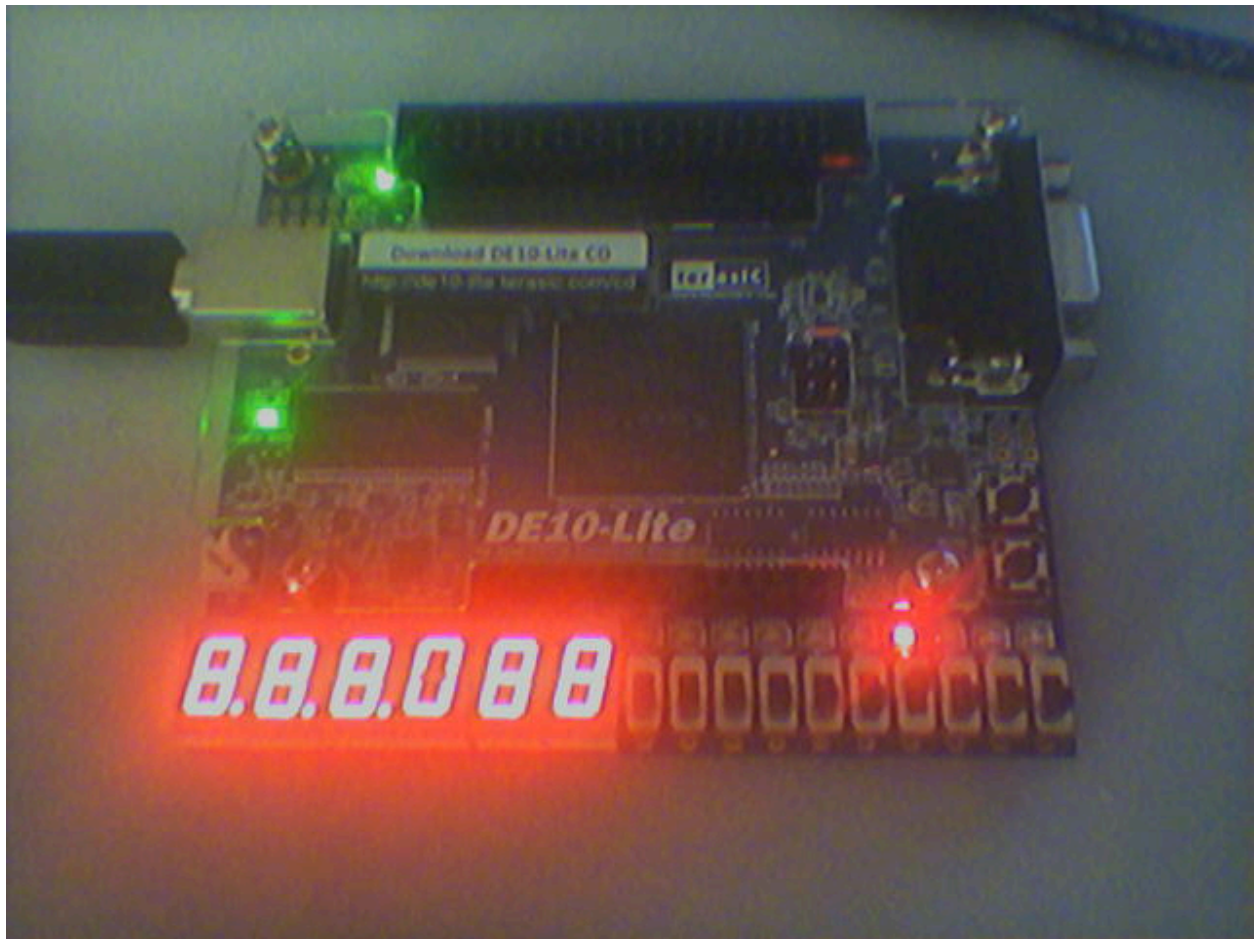


Figure 12: Part 4 Board

## Part 5

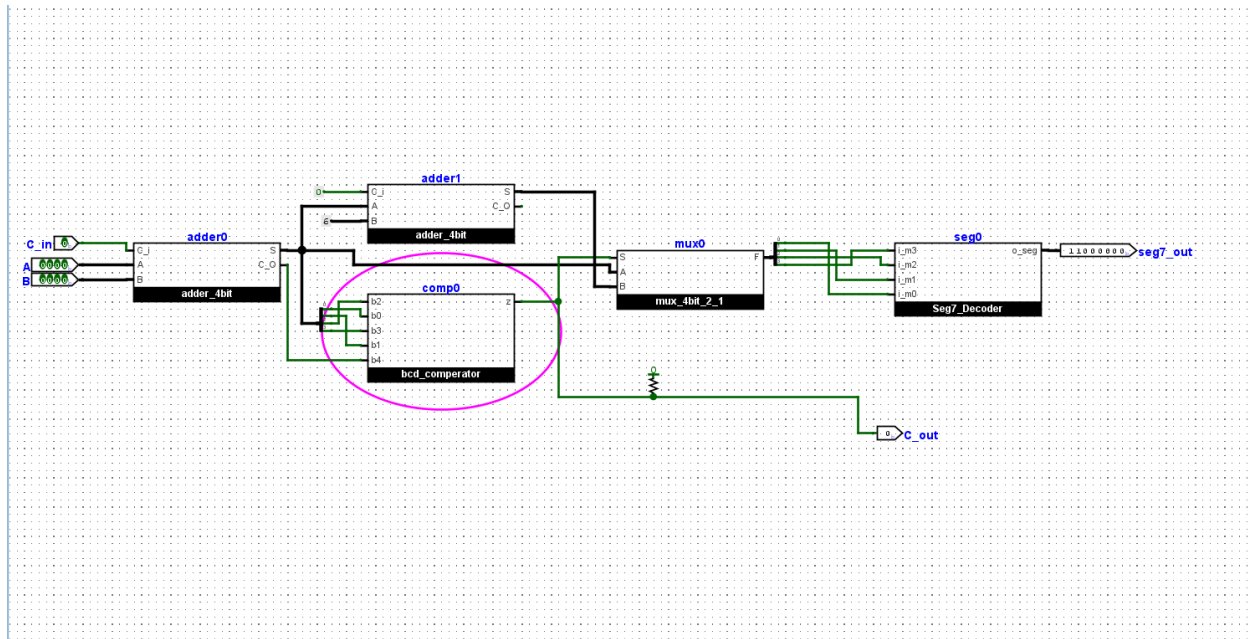


Figure 13: BCD Adder

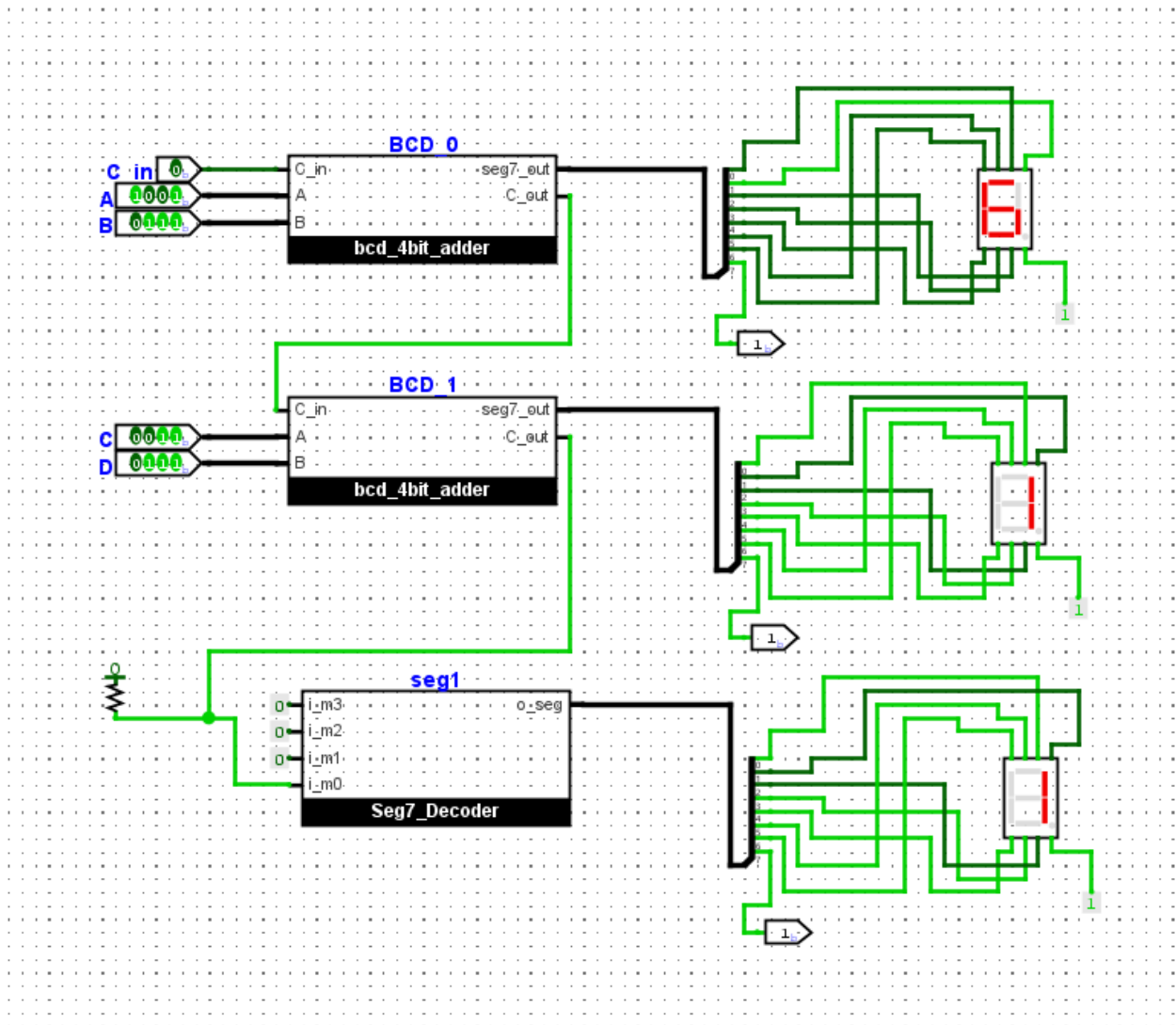


Figure 14: 8Bit BCD Adder

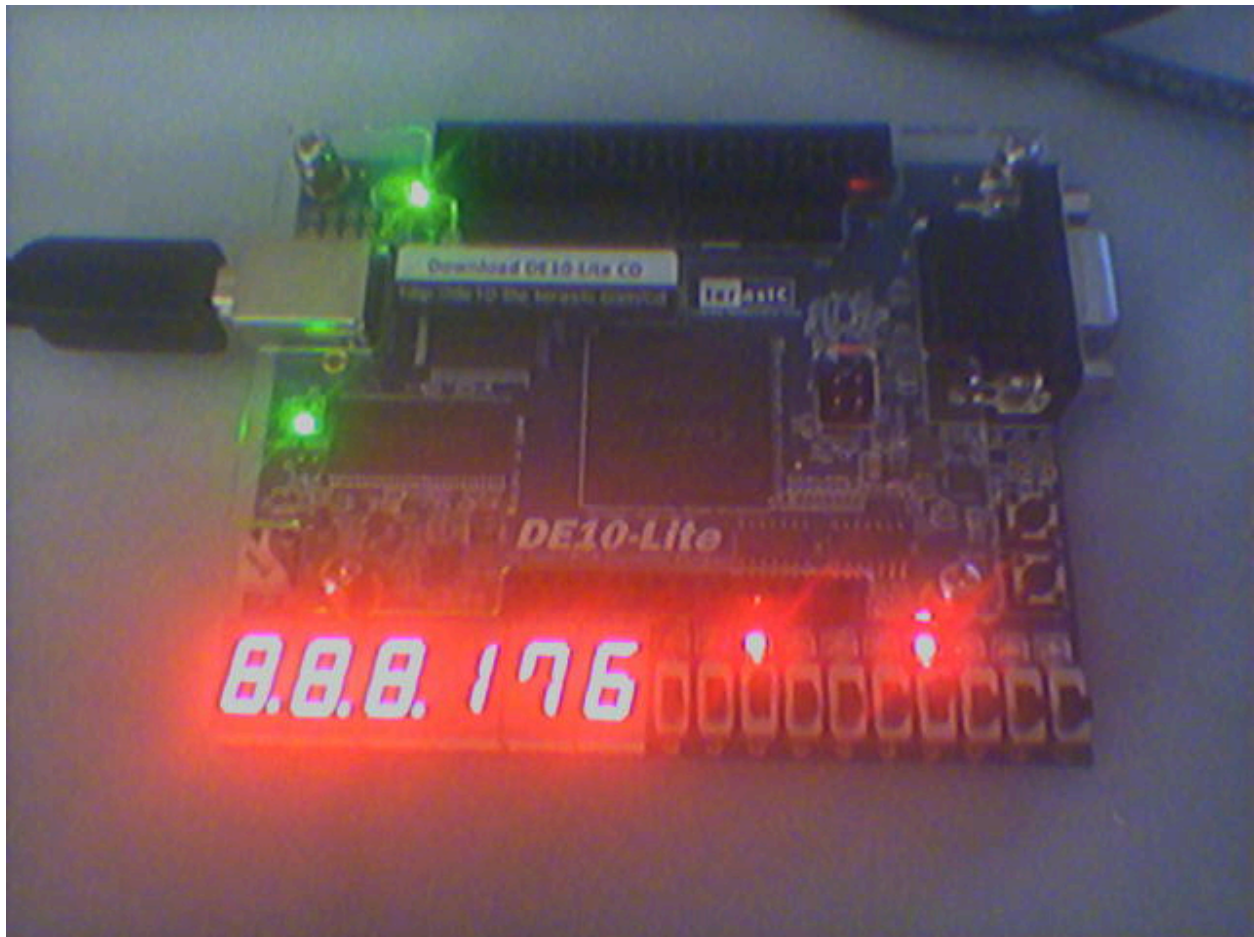


Figure 15: Part 5 Board. Due to Shortage of switches. I made one switch turn on two bits at a time.

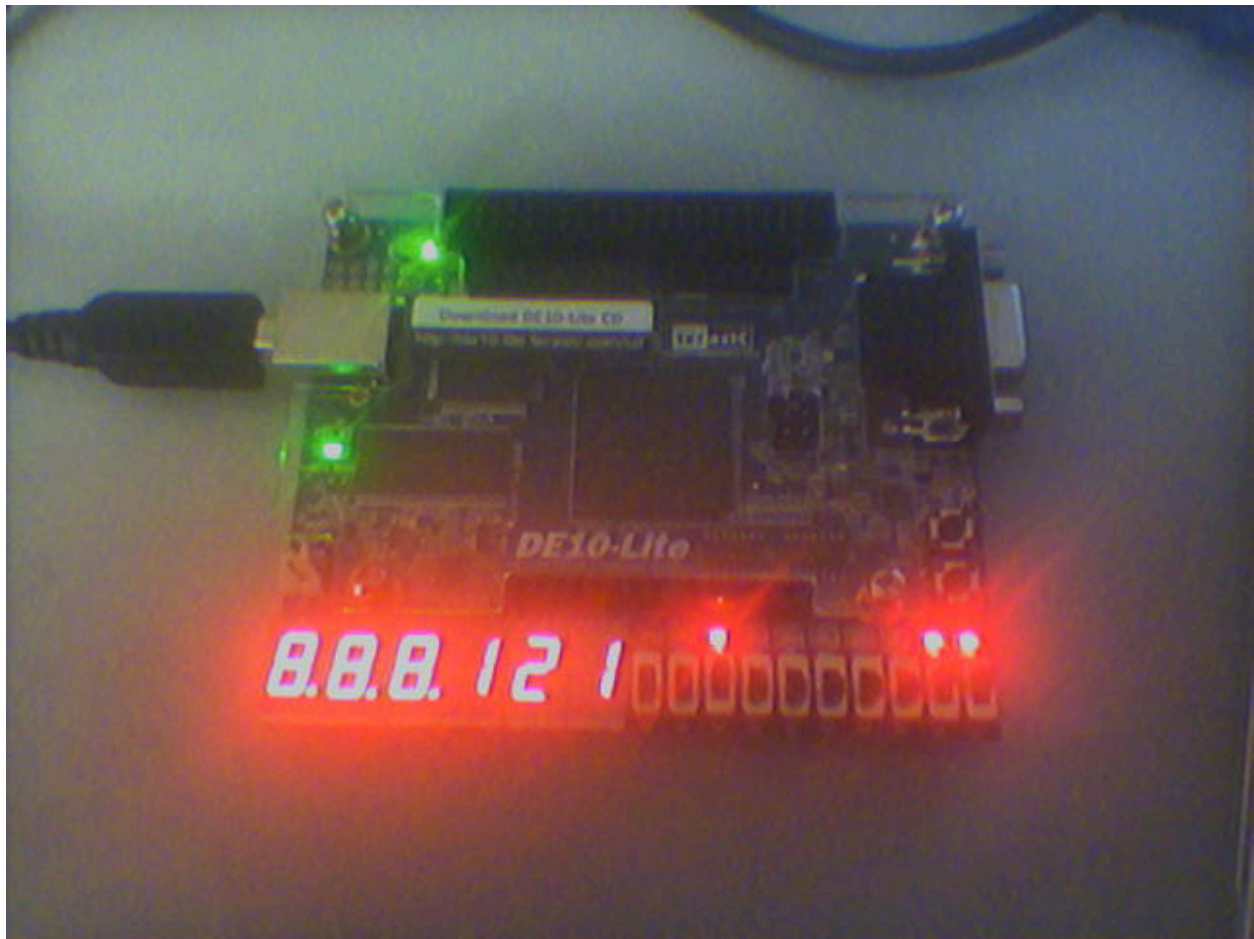


Figure 16: Part 5 Board

## Part 6

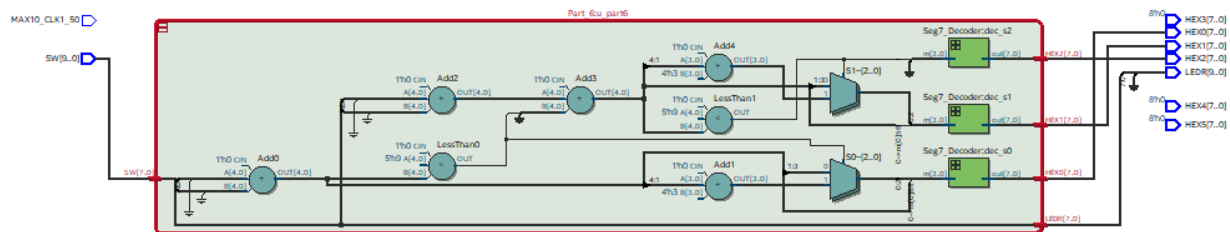


Figure 17: Implementing the same circuit at part 5 with algorithm yields similar design with different implementation. The compiler allows multiple ways to come to the same function.

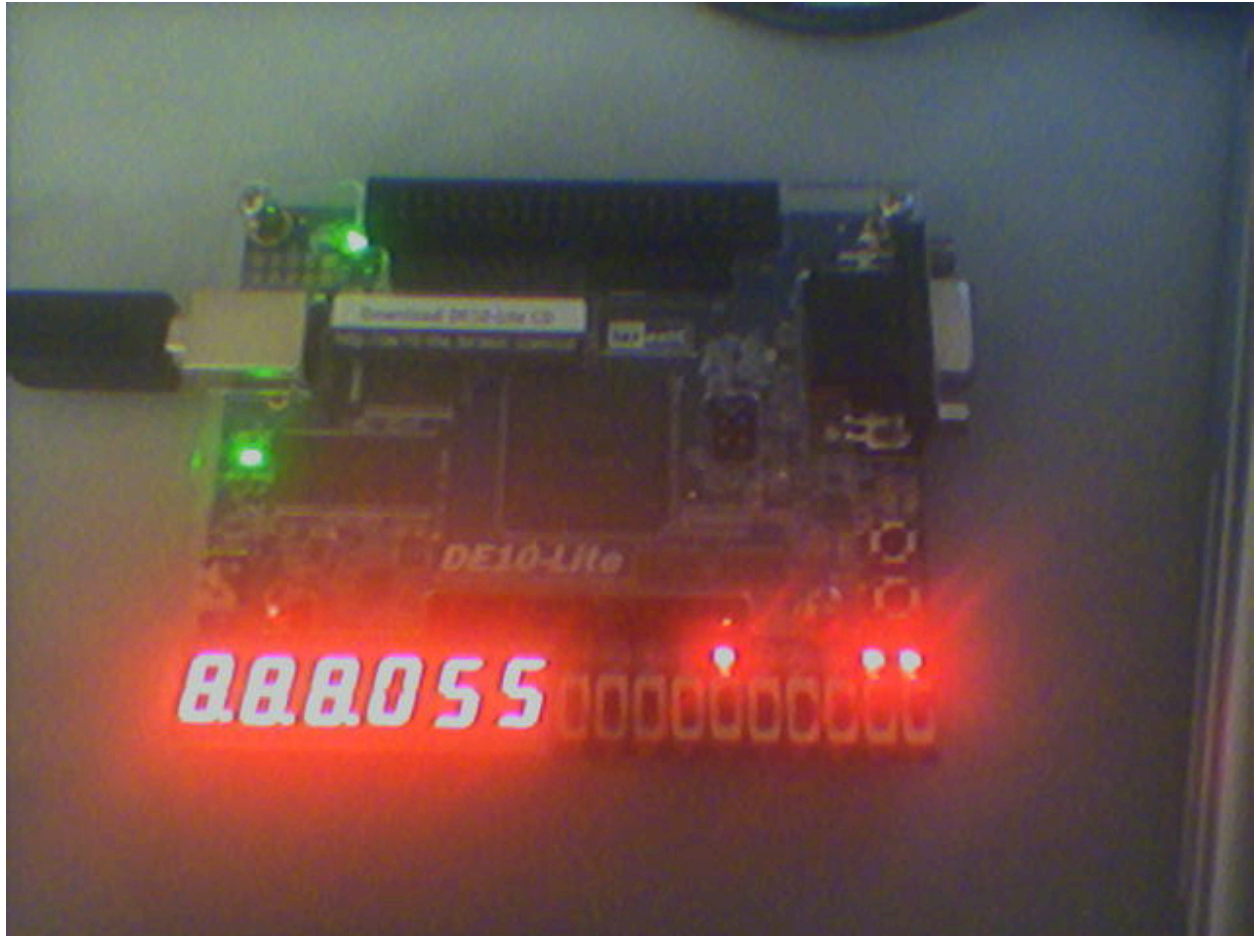


Figure 18: Part 6 Board

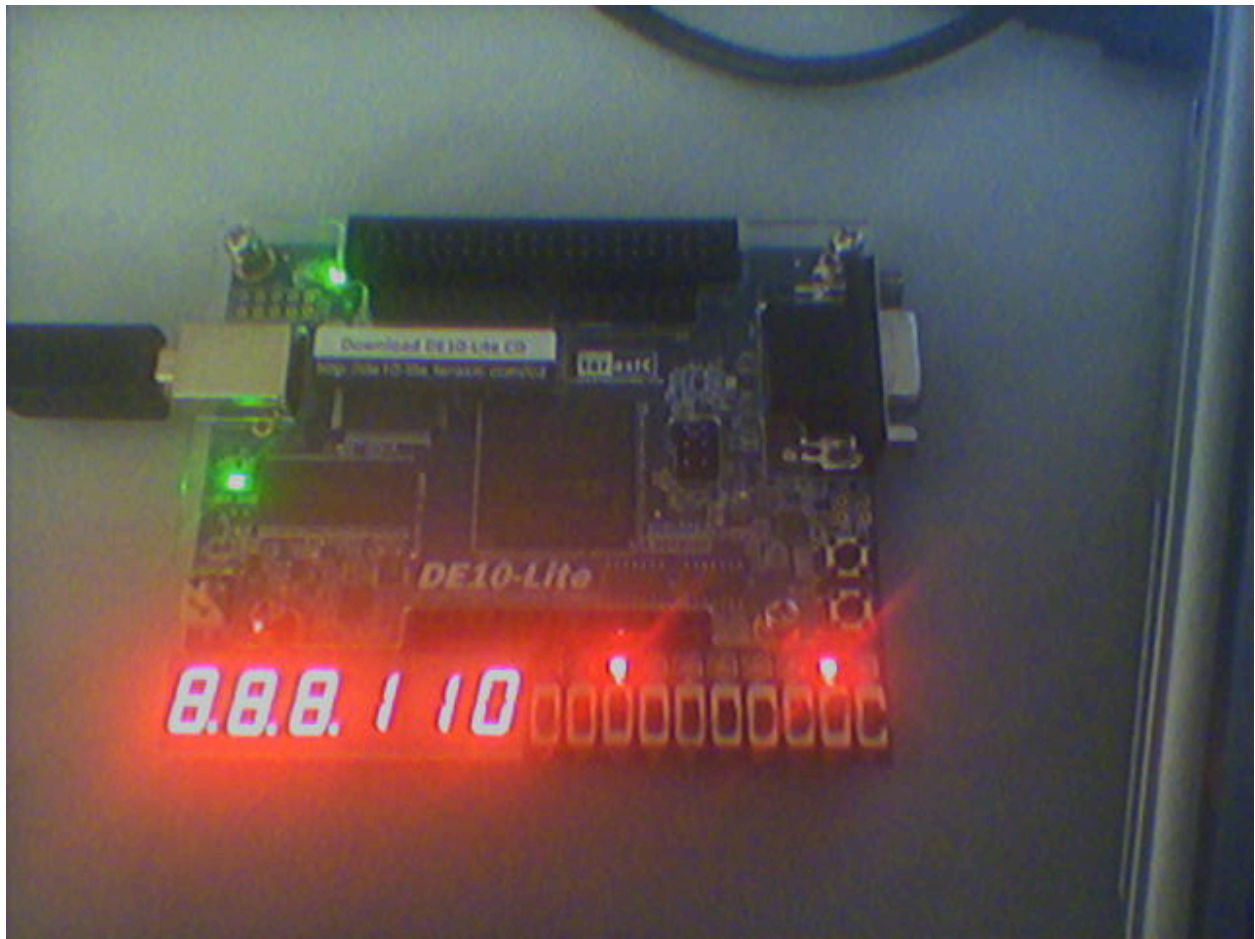


Figure 19: Part 5 Board

## Part 7

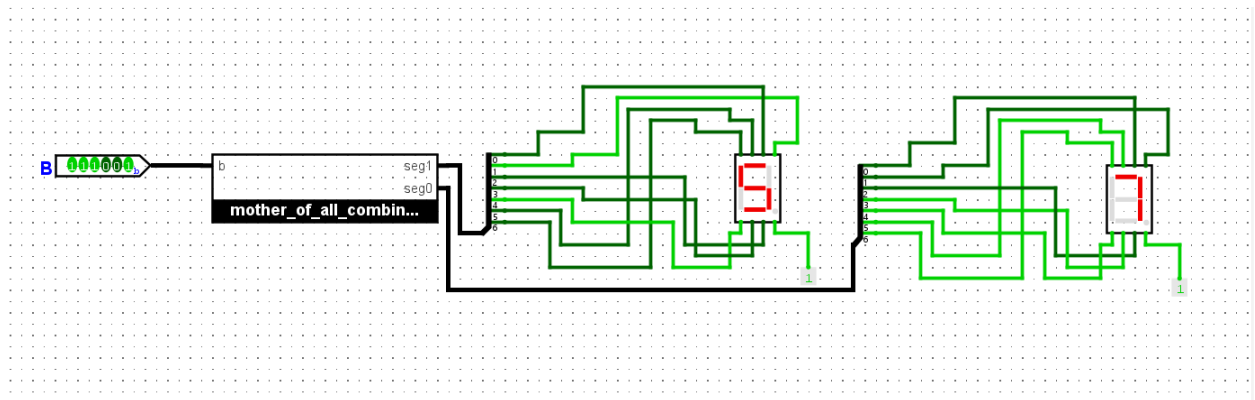


Figure 20: The Mother of All Combinational Circuits

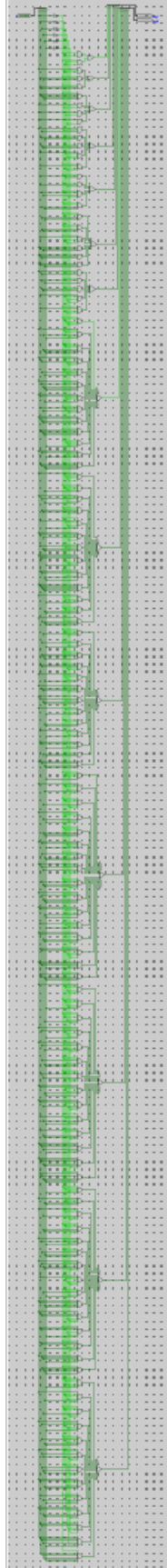


Figure 21: Used Combinational Analysis and the entire Truth table going from binary 6Bits to 7Segment output.

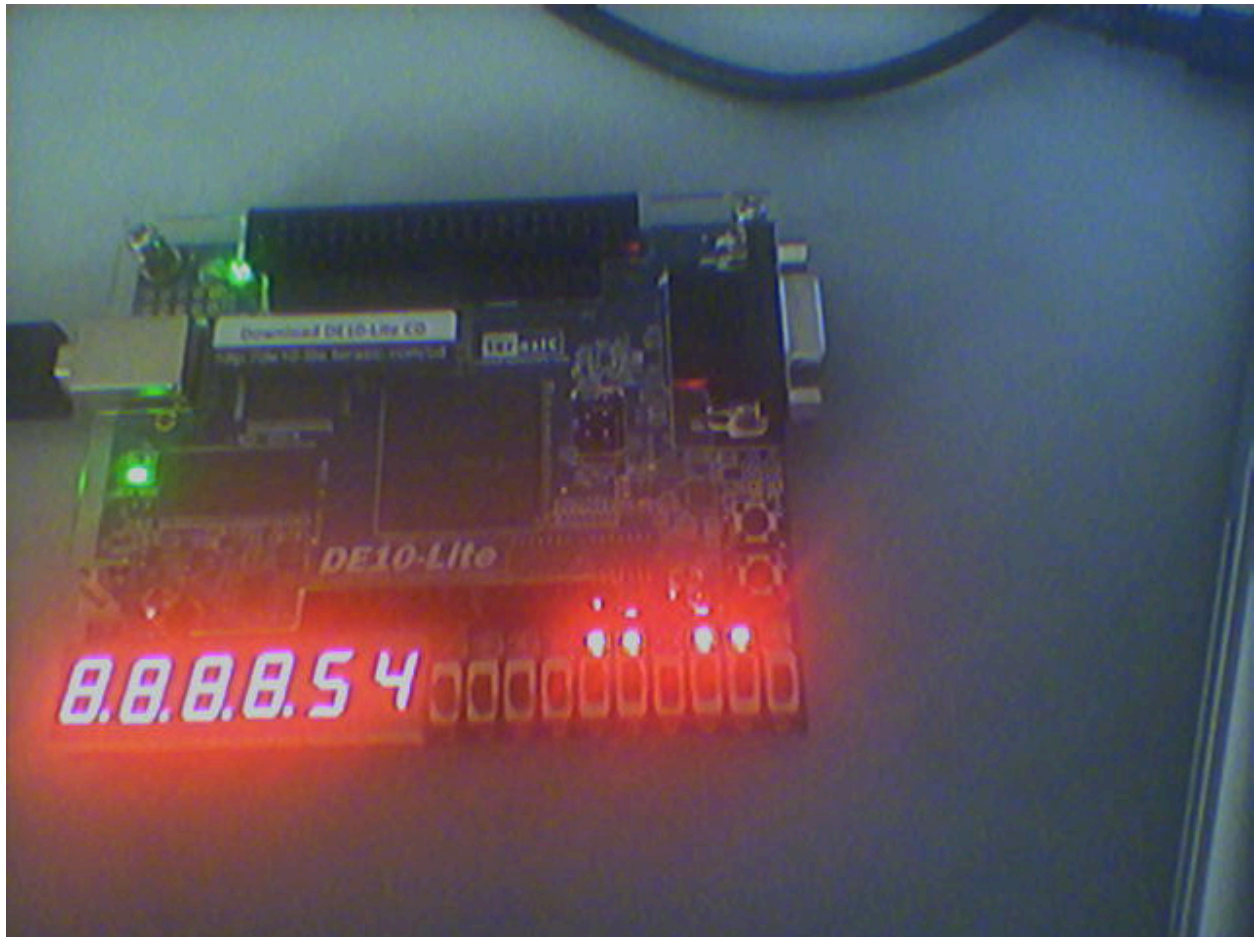


Figure 22: Part 7 Board

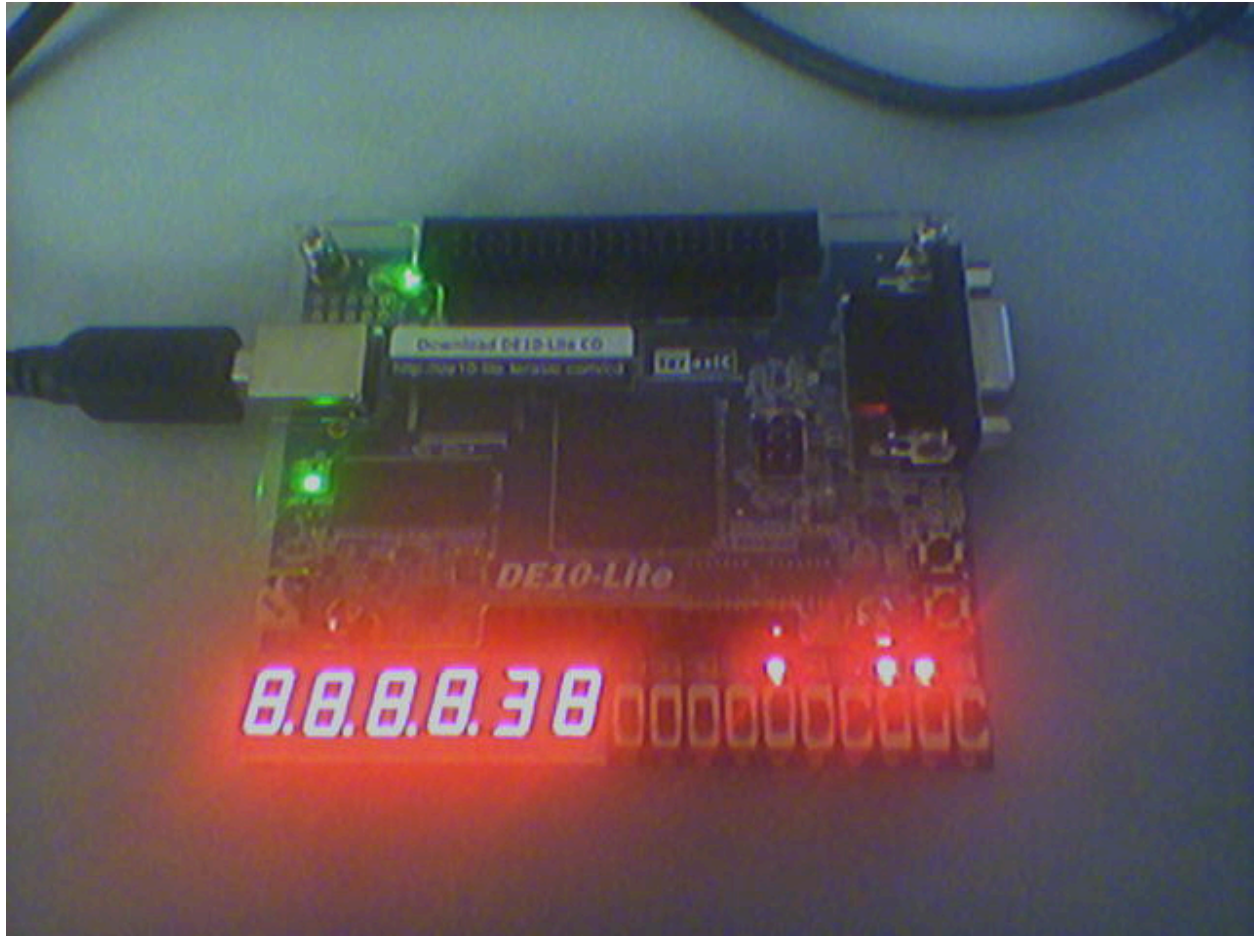


Figure 23: Part 7 Board